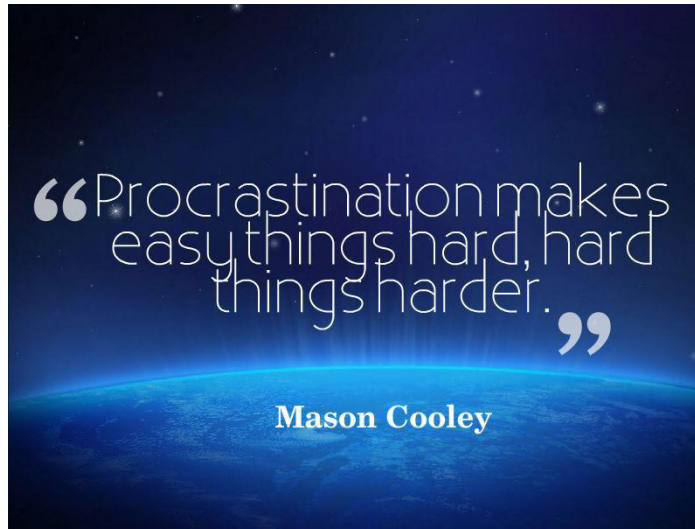


Bits I



Good
Morning

Content

01. Truth Table
02. Basic AND OR XOR properties
03. Left & Right shift operator
04. check i^{th} bit
05. Count set bits
06. Unset i^{th} bit
07. set bits in Range

Very
Easy

Truth Table

A	B	$A \& B$	$A B$	$A \wedge B$	$\sim A$
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

$A \& B \Rightarrow$ Both the bits has to be set

$A | B \Rightarrow$ Any bit is set

$A \wedge B =$ Both the bits has to be different

* Basic AND properties

01. $A \& 1 \begin{cases} 0 \text{ (even)} \\ 1 \text{ (odd)} \end{cases}$

$A = 10$

$$\begin{array}{r} 1010 \\ \& 0001 \\ \hline 0000 \end{array}$$

$A = 11$

$$\begin{array}{r} 1011 \\ \& 0001 \\ \hline 0001 \end{array}$$

02. $A \& 0 = 0$

03. $A \& A = A$

* Basic OR properties

$$A \mid 0 = A$$

$$A \mid A = A$$

$$A \mid 1 \begin{cases} \rightarrow A \text{ (odd)} \\ \rightarrow A+1 \text{ (even)} \end{cases}$$

$A = 10$

$$\begin{array}{r} 1010 \\ 10001 \\ \hline 1011 \text{ (A+1)} \end{array}$$

$$A = 11$$

$$\begin{array}{r} 1011 \\ 10001 \\ \hline 1011 \text{ (A)} \end{array}$$

* Basic XOR properties

01. $A \wedge 0 = A$

02. $A \wedge A = 0$

* Commutative property

$$A \& B = B \& A$$

$$A \mid B = B \mid A$$

$$A \wedge B = B \wedge A$$

* Associative property

$$01. \quad A \wedge B \wedge C = (A \wedge B) \wedge C = A \wedge (B \wedge C) = (A \wedge C) \wedge B$$

$$02. \quad A \& B \& C = (A \& B) \& C$$

$$03. \quad A | B | C = (A | B) | C$$

$$* \quad a \wedge b \wedge a \wedge d \wedge b$$

$$= a \wedge a \wedge b \wedge b \wedge d \Rightarrow 0 \wedge 0 \wedge d \Rightarrow d$$

$$* \quad 1 \wedge 3 \wedge 5 \wedge 3 \wedge 2 \wedge 1 \wedge 5$$

$$= \underbrace{1 \wedge 1}_{0 \wedge 2} \wedge \underbrace{2 \wedge 3 \wedge 3}_{0 \wedge 0} \wedge \underbrace{5 \wedge 5}_{0 \wedge 0} \Rightarrow \underline{\underline{2}}$$

* Left shift operator

$$A = 10 \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad = A = 10 = A * 2^0$$

$$A \ll 1 \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{0} \quad = 20 = A * 2^1$$

$A \ll 2$ $\downarrow$ γ $\underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{0} \quad \underline{0} = 40 = A * 2^2$
 $A \ll 3$ $\downarrow$ $\times$ $\underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} = 80 = A * 2^3$
 $A \ll 4$ $\downarrow$ $\times$ $\underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} = 160 = A * 2^4$
 $A \ll 5$ $\downarrow$ $\times$ $\underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} = \text{Ideal case}$
 $\downarrow$
320 X

Overflow issue

$$a \ll n = a * 2^n$$

$$1 \ll n = 1 * 2^n$$

* Right shift operator

$A = 10$ $\underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} \Rightarrow A \gg 0 = 10$

$A \gg 1$ $\underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \times = A \gg 1 = 5$

$A \gg 2$ $\underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \times \Rightarrow A \gg 2 = 2$

$$a \gg n = \frac{a}{2^n}$$

$$* 1 \ll 3 \Rightarrow 1 * 2^3 \Rightarrow 8$$

* Power of left shift operator

$$01. (45)_{10} = (101101)_2$$

$$\begin{array}{r} 45 = 101101 \\ \& 1 \ll 2 = 000100 \\ \hline 000100 \end{array}$$

$$\begin{array}{r} 45 = 101101 \\ \& (1 \ll 4) = 010000 \\ \hline 000000 \end{array}$$

$$\begin{array}{r} 45 = 101101 \\ \& 1 \ll 1 = 000010 \\ \hline 000010 \end{array}$$

$N \& (1 \ll i)$ $\begin{cases} 0 & (i^{\text{th}} \text{ bit is unset}) \\ (1 \ll i) & (i^{\text{th}} \text{ bit is set}) \end{cases}$

* Power of left shift with OR

OR

$$\begin{array}{r} 45 = 101101 \\ 1 \ll 2 = 000100 \\ \hline 101101 \end{array}$$

OR

$$\begin{array}{r} 45 = 101101 \\ 1 \ll 4 = 010000 \\ \hline 111101 \end{array}$$

OR

$$\begin{array}{r} 45 = 101101 \\ 1 \ll 1 = 000010 \\ \hline 101111 \end{array}$$

$N \mid (1 \ll i) \longrightarrow$ set the i bit in ans

* left shift with xor operator

$$\begin{array}{r} 45 = 101101 \\ \wedge \\ 1 \ll 2 = 000100 \\ \hline 101001 \end{array}$$

$$\begin{array}{r} 45 = 101101 \\ \wedge \\ 1 \ll 4 = 010000 \\ \hline 111101 \end{array}$$

$$\begin{array}{r} 45 = 101101 \\ \wedge \\ 1 \ll 1 = 000010 \\ \hline 101111 \end{array}$$

$N \wedge (1 \ll i) \rightarrow$ toggle my i^{th} bit

7:54 AM $\rightarrow$ 8:04 AM